A prime-five smooth Cantor cycle

For $y^2 = x^5+1$ over $\mathbf{F}_3$, Lean proves that the complete reduced polynomial-pair model has ten elements, eight with split u. The actual five pairs $(1,0),(X,\pm1),(X^2,\pm1)$ form a cyclic arithmetic image. Its generator $(X,1)$ has order five, different from the characteristic. All five members are smooth, giving an exact inequality $1 \neq 4/5$ against the ambient fraction. [pcc:unverified]

The count proof classifies every one of the 91 bounded coefficient shapes and proves a bijection with arbitrary actual pairs: u monic, $\deg u \leq 2$, $\deg v < \deg u$, $u \mid v^2 - (X^5+1)$. It proves exactly ten valid pairs. Exactly two have nonsplit $u = X^2+X+2$, with $v = \pm X$; splitting is the actual `Polynomial.Splits` predicate. The curve polynomial is monic, degree five and separable. [pcc:unverified]

For every ordered pair in the five-cycle, an explicit certificate gives common divisors and a Bezout equation identifying the actual normalized full gcd. The proof then checks the composition and numerator equations:

$$d = \gcd(\gcd(u_P, u_Q), v_P + v_Q), \qquad d^2U = u_P u_Q,$$
$$au_P + bu_Q + c(v_P + v_Q) = d,$$
$$au_P v_Q + bu_Q v_P + c(v_P v_Q + f) = d(V + Uq).$$

The monic intermediate U, reduced intermediate V, and exact final reduction are checked in all 25 cases. Only after this matching is the Fin5 group transported to the actual five pairs. Lean proves generation, order five, properness of the image and both exact fractions. [pcc:unverified]

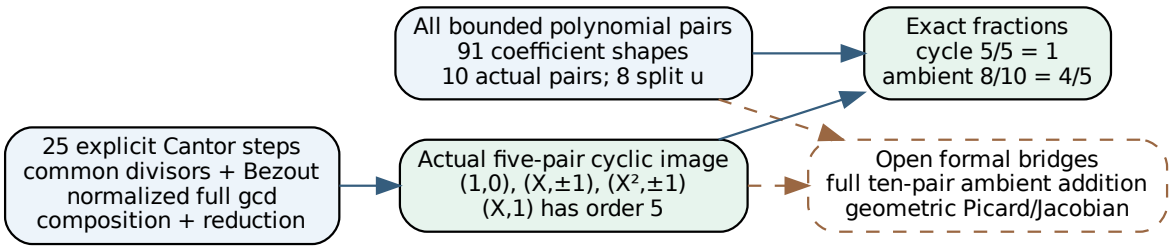


Figure 1: Actual polynomial classification and all 25 addition certificates establish the finite cyclic-image inequality. Dashed bridges remain open.

The full ten-pair ambient group operation and geometric Picard/Jacobian identification remain separate formal obligations. This proves a finite arithmetic cycle inequality; it does not refute Gaudry's fixed-genus limit or a qualified large-prime/typical-subgroup heuristic. [pcc:unverified]

Proof map and verification scope

The combined declaration is

`MumfordStollCycleGroupAudit.stoll_cycle_audit_certificate`.

Obligation	Declaration, abbreviated namespace
Arbitrary-pair completeness	Stoll.canonical_valid_shape
Complete ten/eight counts	Stoll.reducedPair_cardinality / smooth_reducedPair_cardinality
All 25 actual additions	StollCantor.cycle_all_pairs
Actual normalized full gcd	CycleGroup.normalized_full_gcd / step_has_actual_full_gcd
Final remainder	CycleGroup.reduction_matches_remainder
Group addition matches certificates	CycleGroup.addition_is_Cantor
Generator/order/generation	generator_coordinates / generator_order / generated_by_actual_point
Properness and unequal fractions	cycle_is_proper / exact_proportion_inequality

Exact full names and hypotheses are in `../../STOLL_CYCLE.md` and the three Lean modules. The tracked project gate passes 348 declarations, zero failures and zero trusted-native exceptions, including generated declarations. Lean 4.35.0 and Mathlib revision `56a005855ea17e0b0958f73ec801ae140b0a1679` are pinned. Only propext, Classical.choice and Quot.sound are accepted; no holes, custom mathematical axioms or native evaluation are used. [pcc:unverified]

Source/plan commit `327d3183` precedes supplement `IDC1hb7102c15384f2648` and theorem packet `PCC-20261109-d8b77b`. The local PCC check passes at 23:00:19Z on producer host `762f5fc120c9`. Signed standing remains unverified. Source/issuer/lock bytes were checked against the commit before issue and again afterward. [pcc:unverified]

The scalar supplement checks 32 derived and 11 extra seeded points over $2^{61}-1$, for five variables and total degree five. It covers only the quadratic quotient/remainder rewrite for x^5+1. Its exact 4117 bytes are retained without a JSON roundtrip. Counts, splitting, actual gcd, all 25 additions, order, generation and fraction inequality rest on Lean; the supplement does not prove those statements. [pcc:unverified]

Failed and warning-bearing attempts remain in a lossless archive with per-file hashes. The final Cantor build and group build pass with no printed warnings; the whole project retains existing Lake module-system advisories. No linter is waived. See `../../stoll-verification.json` and `../../evidence/stoll-cycle-gate.log`. [pcc:unverified]

Primary source: Stoll2014 Example 4.19, printed pp. 25–26, read in full, and the representation/addition sections in `../../stoll-sources.json`. The rational worked example motivates the cycle; characteristic-three validity, completeness and arithmetic are separately checked here. Original Cantor1987 remains unread after publisher403. No novelty claim is made. Native full-population evidence is a separate report. The original fourteen-item audit remains partial. [pcc:unverified]